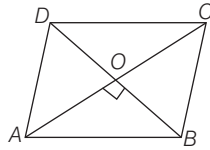


5. (d) Two parallelograms on the same base and between the same parallels are equal in area. So, the ratio of their areas is 1 : 1.

6. (a) As $ABCD$ is a rhombus.

So, $AO = OC = \frac{1}{2} AC$



$$BO = OD = \frac{1}{2} BD, \angle AOB = 90^\circ$$

$$\therefore AB^2 = OA^2 + OB^2$$

$$AB^2 = \frac{AC^2}{4} + \frac{BD^2}{4}$$

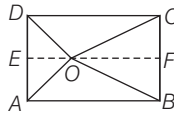
$$\Rightarrow 4AB^2 = AC^2 + BD^2$$

7. (d) Draw $EF \parallel AB$.

In right angled $\triangle OEA$ and $\triangle OFC$,

$$OA^2 = OE^2 + AE^2 \quad \dots (i)$$

and $OC^2 = OF^2 + CF^2 \quad \dots (ii)$



On adding Eqs. (i) and (ii), we get

$$\therefore OA^2 + OC^2 = OE^2 + AE^2 + OF^2 + CF^2 \quad \dots (iii)$$

In right angled $\triangle DEO$ and $\triangle BOF$,

$$OD^2 = OE^2 + DE^2 \quad \dots (iv)$$

$$OB^2 = OF^2 + BF^2 \quad \dots (v)$$

On adding Eqs. (iv) and (v), we get

$$\Rightarrow \text{Area of } \parallel \text{gm } ABCD$$

$$\Rightarrow OD^2 + OB^2 = OE^2 + OF^2 + DE^2 + BF^2 \quad \dots (vi)$$

As $FB = EA$ and $DE = CF$

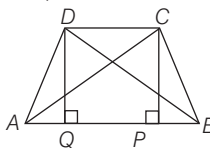
$$\therefore OD^2 + OB^2 = OE^2 + OF^2 + AE^2 + CF^2 \quad \dots (vii)$$

Here, from Eqs. (iii) and (vii), we get

$$OA^2 + OC^2 = OD^2 + OB^2$$

8. (a) Draw $DQ \perp AB$ and $CP \perp AB$.

In $\triangle ABD$, $\angle A$ is acute.



$$\text{So, } BD^2 = AD^2 + AB^2 - 2AB \cdot AQ \quad \dots (i)$$

In $\triangle ABC$, $\angle B$ is acute.

$$\text{So, } AC^2 = BC^2 + AB^2 - 2AB \cdot BP \quad \dots (ii)$$

Adding Eqs. (i) and (ii), we get

$$\therefore AC^2 + BD^2 = (BC^2 + AD^2)$$

$$+ 2AB(AB - BP - AQ)$$

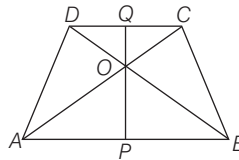
$$= (BC^2 + AD^2) + 2AB \cdot PQ$$

$$[\because PQ = AB - AQ - BP]$$

$$= BC^2 + AD^2 + 2AB \cdot CD$$

$$[\because PQ = DC]$$

9. (c) Let PQ be the perpendicular distance between the two parallel sides AB and CD .
[$\because PO = 2OQ$]



$$\text{So, } \frac{\text{Area of } \triangle AOB}{\text{Area of } \triangle COD} = \frac{\frac{1}{2} AB \times OP}{\frac{1}{2} CD \times OQ}$$

$$= \frac{2CD \times 2OQ}{CD \times OQ} = \frac{4}{1} \text{ or } 4 : 1$$

10. (b) Area of parallelogram $ABCD$
 $= 2 (\text{Area of } \triangle ADE)$

$$+ (\text{Area of rectangle } AECH)$$

$$= 2 \times \frac{1}{2} DE \times AE + AE \times EC$$

$$= DE \times AE + AE \times EC$$

$$= AE(DE + EC) = AE \times [DC]$$

$$\therefore \frac{DE}{EC} = \frac{1}{2} \Rightarrow \frac{DE}{EC} + 1 = \frac{1}{2} + 1$$

$$\Rightarrow \frac{DE + EC}{EC} = \frac{3}{2} \Rightarrow \frac{DC}{EC} = \frac{3}{2}$$

$$\text{Area of parallelogram } ABCD = \frac{FE}{3} \times \frac{3EC}{2} = \frac{FE \times EC}{2}$$

$$[\because FE = 3AE]$$

$$= \frac{\text{Area of rectangle } EFGC}{2}$$

$\therefore$ Required ratio = 1 : 2

11. (a) Join AC and BD .

As $AB > BC$

$$\Rightarrow \angle ACB > \angle BAC \quad \dots (i)$$

Also, $AD > DC$

$$\Rightarrow \angle ACD > \angle CAD \quad \dots (ii)$$

On adding Eqs. (i) and (ii), we get

$$\therefore \angle ACB + \angle ACD > \angle BAC + \angle CAD$$

$$\Rightarrow \angle C > \angle A, \text{ similarly } \angle D > \angle B$$

12. (b) $\angle COD = 105^\circ$

$$\therefore \angle BOA = 105^\circ$$

$$[\text{vertically opposite angles}]$$

$$\therefore \angle AOD = \angle COX = \lambda$$

$$\Rightarrow 105^\circ + 105^\circ + x + x = 360^\circ$$

$$\Rightarrow 2x = 150^\circ$$

$$\Rightarrow x = 75^\circ$$

$$\text{Also, } \angle OCX = \frac{1}{2} \times \angle C = \frac{1}{2} \times 90^\circ = 45^\circ$$

In $\triangle OCX$,

$$\angle OCX + \angle COX + \angle OXC = 180^\circ$$

[by angle sum of property of a triangle]

$$\Rightarrow 45^\circ + 75^\circ + \angle OXC = 180^\circ$$

$$\Rightarrow \angle OXC = 180^\circ - 120^\circ = 60^\circ$$

$$\Rightarrow x = 60^\circ$$

13. (a) $EB \parallel DL$ and $ED \parallel BL$

$\Rightarrow EBLD$ is a parallelogram.

$$\therefore BL = ED = \frac{1}{2} AD = \frac{1}{2} BC = LC$$

Also, in $\triangle DCL$ and $\triangle FBL$,

$$LC = BL, \angle DLC = \angle FLB$$

$\therefore$ [vertically opposite angles]

and $\angle CDL = \angle BFL$ [alternate angles]

$\therefore \triangle DCL \cong \triangle FBL$ [By AAS rule]

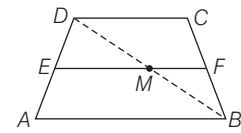
$$\therefore DC = BF \text{ and } DL = FL$$

$$\therefore BF = DC = AB$$

$$\Rightarrow 2AB = 2DC$$

$$\Rightarrow AF = 2DC \quad [\because BF = AB]$$

14. (b) Join BD , cutting EF at M .



So, M is mid-point of BD .

[$\because E$ is mid-point of AD and $EM \parallel AB$]

$$\therefore EM = \frac{1}{2} AB \quad \dots (i)$$

Similarly, F is mid-point of BC in $\triangle BCD$.

$$\therefore FM = \frac{1}{2} DC \quad \dots (ii)$$

$$\therefore EF = EM + MF = \frac{1}{2} (AB + DC)$$

[from Eqs. (i) and (ii)]

15. (a) $\angle CRQ = \angle CBA = 90^\circ \Rightarrow QR \parallel AB$

In $\triangle ABC$, Q is the mid-point of AC and $QR \parallel AB$.

So, R is mid-point of BC .

Similarly, P is the mid-point of DC .

$$\therefore DP = PC$$

16. (d) Number of diagonals of a polygon of

$$8 \text{ sides} = \frac{n(n-1)}{2} - n$$

$$= \frac{8(8-1)}{2} - 8 = 20$$

17. (c) Sum of all angles of a pentagon

$$= (5-2) \times 180^\circ = 3 \times 180^\circ = 540^\circ$$

Let the angle be $x, 2x, 3x, 5x$ and $9x$.

$$\therefore x + 2x + 3x + 5x + 9x = 540^\circ$$

$$20x = 540^\circ \Rightarrow x = 27^\circ$$

$$\therefore \text{Largest angle} = 9x = 9 \times 27^\circ = 243^\circ$$

18. (c) By condition,

$$\frac{\text{Interior angle of a regular polygon}}{\text{Exterior angle of a regular polygon}} = \frac{5}{1}$$

$$\Rightarrow \frac{\frac{(n-2)}{n} \times 180^\circ}{\frac{360^\circ}{n}} = \frac{5}{1} \Rightarrow \frac{(n-2)}{2} = \frac{5}{1}$$

$$\Rightarrow n - 2 = 10 \Rightarrow n = 12$$

19. (b) Here, $n = 6$
 Sum of interior angles of a hexagon
 $= (n-2)180^\circ = (6-2) \times 180^\circ$
 $= 4 \times 180^\circ = 720^\circ$
 Number of angles = 6
 $\therefore$ Each angle of a regular hexagon
 $= \frac{720^\circ}{6} = 120^\circ$

20. (c) Each interior angle of a regular polygon

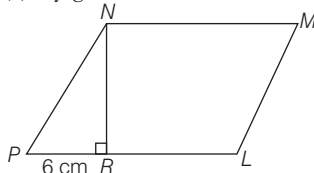
$$= \frac{(n-2) \times 180^\circ}{n}$$

$$\therefore \frac{(n-2) \times 180^\circ}{n} = 150^\circ \quad (\text{given})$$

$$\Rightarrow \frac{(n-2)180}{n} = 150 \Rightarrow 30n = 360 \Rightarrow n = \frac{360}{30}$$

$$\Rightarrow n = 12$$

21. (b) By given condition,



Area of parallelogram $LMNP$
 $= 6 \times \text{Area of } \triangle NRP$

$$\therefore NR \times PL = 6 \times \frac{1}{2} \times NR \times PR$$

$$\Rightarrow PL = 3PR$$

$$\Rightarrow PR + RL = 3PR \quad [\because PL = PR + RL]$$

$$\Rightarrow RL = 2PR = 2 \times 6 = 12 \text{ cm}$$

22. (d) $ABCD$ is a trapezium in which $AB \parallel DC$ and M, N are the mid-points of AD and BC .

Hence, $MN \parallel AB$ and $MN \parallel DC$.

In $\triangle ACB$, ON passes through the mid-point N of BC and $ON \parallel AB$

$$\therefore ON = \frac{1}{2}AB = \frac{1}{2} \times 12 \text{ cm} = 6 \text{ cm}$$

$$\text{And, } MO = MN - ON = (14 - 6) \text{ cm} = 8 \text{ cm}$$

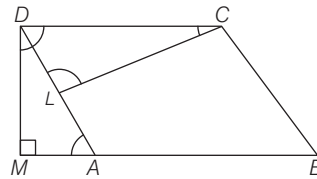
Again, MO passes through the mid-point M of AD and $MO \parallel DC$

$$\therefore MO = \frac{1}{2}DC = \frac{1}{2}CD$$

Hence, $CD = 2(MO) = 2(8) = 16 \text{ cm}$

23. (a) $\triangle ADM \sim \triangle DCL$

Since, $\angle DMA = \angle CLD = 90^\circ$ and $\angle CDL = \angle DAM$ as they are alternate angles



$$(\angle D / \angle C) = (\angle M / \angle L)$$

$$\Rightarrow \frac{AD}{CD} = \frac{DM}{CL} = \frac{12 \times 16}{15} = 12.8 \text{ units}$$

24. (a) Evidently $DP = 14 - 9 = 5 \text{ cm}$

From $\triangle BPC$,

$$BC^2 = 15^2 - 9^2 = 12^2$$

$$\Rightarrow BC = 12 \text{ cm}$$

$$\text{From } \triangle APD, AP^2 = AD^2 + DP^2 = 12^2 + 5^2 = 169$$

$$\Rightarrow AP = 13 \text{ cm}$$

In $\triangle ABP$, $AP < AB < BP$.

Therefore, $\gamma < \beta < \alpha$ (i.e.) $\alpha > \beta > \gamma$

25. (d) Since, $AB = BC$

$$\therefore \angle BAC = \angle BCA = 62^\circ$$

Also as $AB \parallel CD$ and AC transversal

So, $\angle BAC = 62^\circ = \angle ACD$

(alternate interior angles)

$$\therefore \angle ACD = 62^\circ$$

26. (c) One angle of the pentagon is 140° .
 Since, the remaining angles are in the ratio $1 : 2 : 3 : 4$.

$\therefore$ Let the remaining angles be x° , $(2x)^\circ$, $(3x)^\circ$ and $(4x)^\circ$.

But the sum of interior angles of a pentagon

$$(2 \times 5 - 4) \times 90 = 6 \times 90^\circ = 540^\circ$$

$$\therefore 140 + x + 2x + 3x + 4x = 540$$

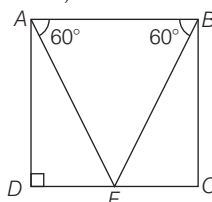
$$\Rightarrow 10x = 400 \Rightarrow x = 40$$

$\therefore$ The angles of the pentagon are 140° , 40° , 80° , 120° and 160° .

Hence, the size of the greatest angle
 $= 160^\circ$

27. (a) Given that, $AB \parallel DC$ and $AD \parallel BC$

In $\triangle ABE$,



$$\Rightarrow \angle EAB = \angle ABE = 60^\circ$$

So, $\triangle ABE$ is an equilateral triangle.

Now, perimeter of $\triangle ABE = 6$

$$\Rightarrow AB + BE + EA = 6$$

$$\Rightarrow AB = 2 \text{ units}$$

Now, In $\triangle ADE$ and $\triangle BCE$,

$$\angle ADE = \angle BCE \quad [\text{each } 90^\circ]$$

$$\angle DAC = \angle CBE \quad [\text{each } 30^\circ]$$

and $AE = BE$

$$\therefore \triangle ADE \cong \triangle BCE \quad [\text{By AAS}]$$

$$\Rightarrow DE = CE \quad [\text{By CPCT}]$$

In $\triangle ADE$ by pythagoras theorem,

$$AE^2 = AD^2 + ED^2$$

$$\Rightarrow 4 = AD^2 + 1$$

$$(\because E \text{ is mid-point of } CD)$$

$$\Rightarrow AD = \sqrt{3} \text{ units}$$

Hence, area of quadrilateral

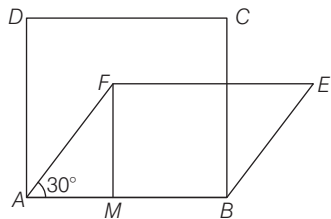
$$ABCD = AB \times AD$$

$$= 2 \times \sqrt{3} = 2\sqrt{3} \text{ sq units}$$

28. (b) $ABCD$ is square and $ABEF$ is a rhombus.

$$\frac{FM}{AF} = \sin 30^\circ = \frac{1}{2}$$

$$\therefore FM = \frac{AF}{2}, AF = AB$$



$$\text{Area of square} = a^2 \quad (AB = AD = a)$$

$$\text{Area of rhombus} = \frac{a \times a}{2} \quad \left(FM = \frac{a}{2} \right)$$

(Area of rhombus = base $\times$ height)

$$\therefore \frac{\text{Area of square}}{\text{Area of rhombus}} = \frac{2}{1}$$

29. (c) In $\triangle ARB$, P is the mid-point of AB and $PD \parallel BR$.

$$\Rightarrow D \text{ is the mid-point of } AR.$$

$\therefore ABCD$ is a parallelogram.

$$\Rightarrow DC \parallel AB \Rightarrow DQ \parallel AB$$

Thus, in $\triangle ARB$, D is the mid-point of AR and $DQ \parallel AB$.

$$\therefore Q \text{ is the mid-point of } RB \Rightarrow BR = 2BQ$$

30. (a) In $\triangle BDC$, Q and R are the mid-points of BD and CD respectively.

$$\therefore QR \parallel BC \text{ and } QR = \frac{1}{2}BC$$

$$\text{Similarly, } PS \parallel BC \text{ and } PS = \frac{1}{2}BC$$

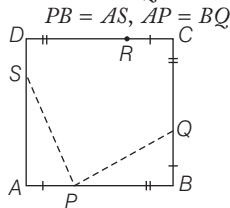
$$\therefore PS \parallel QR \text{ and } PS = QR$$

$$\left[\text{each equal to } \frac{1}{2}BC \right]$$

Similarly, $PQ \parallel SR$ and $PQ = SR$
 [each equal to $1/2 AD$]
 $\therefore PS = QR = SR = PQ$ [$\because AD = BC$]
 Hence, $PQRS$ is a rhombus.

31. (b) $AB \parallel DC$ and EC is a transversal
 $\Rightarrow \angle BEC = \angle ECD$
 $\Rightarrow \angle BEC = \angle ECB$ [$\because \angle ECD = \angle ECB$]
 $\Rightarrow EB = BC \Rightarrow AE = AD$
 Now, $AE = AD \Rightarrow \angle ADE = \angle AED$
 $\Rightarrow \angle ADE = \angle EDC$
 [$\because$ alternate Int. angles]
 $\therefore DE$ bisects $\angle ADC$
 Again, $\angle ADC + \angle BCD = 180^\circ$
 [Co. Int. angles]
 $\Rightarrow \frac{1}{2} \angle ADC + \frac{1}{2} \angle BCD = 90^\circ$
 $\Rightarrow \angle EDC + \angle DCE = 90^\circ$
 But, $\angle EDC + \angle DEC + \angle DCE = 180^\circ$
 [$\because$ sum of the $\angle$ s of a Δ is 180°]
 $\therefore \angle DEC = 180^\circ - 90^\circ$
 $\therefore \angle DEC = 90^\circ$

32. (d) In ΔAPS and ΔBPQ , [given]



and $\angle A = \angle B = 90^\circ$
 $\therefore \Delta APS \cong \Delta BPQ$ (by SAS rule)
 $\therefore SP = PQ, \angle SPA = \angle BQP$
 and $\angle ASP = \angle BPQ$
 $\angle SPQ = 180^\circ - \angle APS - \angle BPQ$
 $= 180^\circ - (\angle APS + \angle ASP)$
 $= 180^\circ - 90^\circ = 90^\circ$

33. (c) As $ABCD$ is a $\parallel$ gm

$\therefore AB \parallel DC$ and $AB = DC$

and $\frac{1}{2} AB = \frac{1}{2} DC$

$\Rightarrow AP = QC$

$\therefore APCQ$ is a $\parallel$ gm

$\Rightarrow AQ \parallel PC$

In $\Delta BAY, XP \parallel AY$

and P is the mid-point of AB

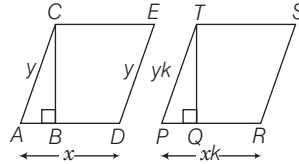
$\therefore BX = YX$

Similarly, in $\Delta DXC, DY = YX$

$\therefore BX = XY = DY$

34. (c) If a parallelogram and a rectangle stand on the same base and on the same side of the base with the same height, then perimeter of parallelogram is greater than perimeter of rectangle.
 $\therefore l_1 > l_2$

35. (c) Let the sides of two parallelograms are x, y and xk, yk , respectively.



Since, sides of two parallelogram are in the ratio $1 : k$.

$\therefore \Delta ABC \sim \Delta PQT$

$\therefore \frac{AC}{PT} = \frac{BC}{QT}$

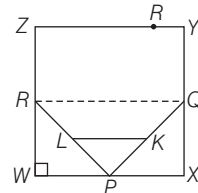
$\Rightarrow \frac{BC}{QT} = \frac{y}{yk} = \frac{1}{k}$

Let $BC = z$ and $QT = zk$

$\therefore$ Ratio of areas of two similar

parallelograms $= \frac{x \times z}{xk \times zk} = \frac{1}{k^2}$ or $1 : k^2$

36. (b) Area of $(\Delta PRQ) = \frac{1}{2}$ Area of $(WXYZ)$



$$= \frac{1}{2} \left[\frac{1}{2} \text{Area of } (WXYZ) \right]$$

$$= \frac{1}{4} \text{Area of } (WXYZ) \quad \dots(i)$$

$$\frac{\text{Area of } (\Delta PRQ)}{\text{Area of } (\Delta PLK)} = \frac{RP^2}{LP^2}$$

(by properties of similar triangle)

$$\Rightarrow \frac{\text{Area of } (\Delta PRQ)}{\text{Area of } (\Delta PLK)} = \frac{(2LP)^2}{LP^2}$$

[$\because L$ and K are the mid-point of PR and PQ]

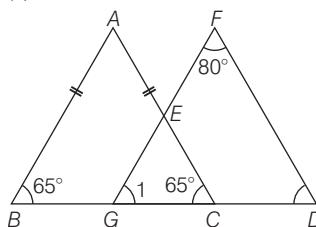
$$\Rightarrow \text{Area of } (\Delta PRQ) = 4 \text{Area of } (\Delta PLK)$$

$$\Rightarrow \frac{1}{4} \text{Area of } (WXYZ)$$

$$= 4 \text{Area of } (\Delta PLK) \text{ [from Eq. (i)]}$$

$$\Rightarrow \frac{\text{Area of } (\Delta PLK)}{\text{Area of } (WXYZ)} = \frac{1}{16}$$

37. (d) Here, $\angle B = \angle C = 65^\circ$ [$\because AB = AC$]



$\angle 1 = \angle B = 65^\circ$ (corresponding angles)

In ΔFGD ,

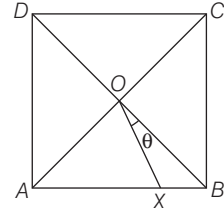
$$\angle 1 + \angle F + \angle D = 180^\circ$$

(by angle sum of property of a triangle)

$$\Rightarrow 65^\circ + 80^\circ + \angle D = 180^\circ$$

$$\Rightarrow \angle D = 35^\circ$$

38. (a) Let $\angle XO B = \theta$



In ΔOXB ,

$$\angle XO B + \angle OBX + \angle OXB = 180^\circ$$

(by angle sum of property of a triangle)

$$\Rightarrow \theta + 45^\circ + \angle OXB = 180^\circ$$

$$\Rightarrow \angle OXB = 180^\circ - 45^\circ - \theta = 135^\circ - \theta$$

$$\text{Here, } \angle OXA + \angle OXB = 180^\circ$$

(linear pair)

$$\Rightarrow \angle OXA + 135^\circ - \theta = 180^\circ$$

$$\Rightarrow \angle OXA = 45^\circ + \theta$$

In ΔOXA , $AO = AX$ (given)

$$\therefore \angle OXA = \angle AOX = 45^\circ + \theta$$

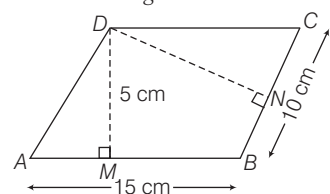
Since, $\angle AOX + \angle XO B = 90^\circ$

$$\Rightarrow 45^\circ + \theta + \theta = 90^\circ$$

$$\Rightarrow 2\theta = 45^\circ \Rightarrow \theta = 22.5^\circ$$

39. (b) Area of parallelogram

$$= \text{Base} \times \text{height} = 15 \times 5 = 75 \text{ cm}^2$$



$$\text{Area of parallelogram} = \text{Base} \times \text{Height}$$

$$= 10 \times DN$$

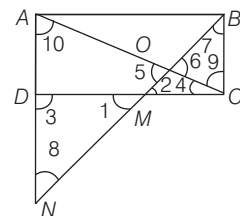
$$\therefore 10 \times DN = 75$$

$$\Rightarrow DN = \frac{75}{10} = 7.5 \text{ cm}$$

40. (b) In ΔDMN and ΔBMC ,

$$DM = MC$$

(given)



$\angle 1 = \angle 2$ (vertically opposite angle)

$$\angle 3 = \angle 4 + \angle 9$$

(alternate interior angle)

$$\Delta DMN \cong \Delta BMC \text{ (as ASA rule)}$$

$$DN = BC = AD \quad (\text{by CPCT})$$

$$\text{So, } AN = 2BC \Rightarrow \frac{AN}{BC} = \frac{2}{1} \dots(i)$$

In $\triangle OAN$ and $\triangle OBC$,

$$\angle 5 = \angle 6 \quad (\text{vertically opposite angle})$$

$$\angle 7 = \angle 8 \quad (\text{alternate interior angle})$$

$$\angle 9 = \angle 10 \quad (\text{altenate angle})$$

$\therefore \triangle OAN \sim \triangle OBC$ (by AAA similarity)

So, the sides will be in same ratio

$$\frac{AN}{BC} = \frac{ON}{OB} \Rightarrow \frac{2}{1} = \frac{ON}{OB} \text{ or } 2 : 1$$

[from Eq. (i)]

41. (a) As $ABCD$ is a parallelogram.

$$\therefore AB \parallel CD \text{ and } AB = CD$$

Draw $QR \perp AB$

In $\triangle DOQ$ and $\triangle POB$,

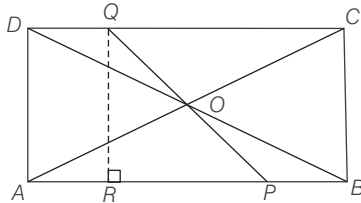
$$\angle ODQ = \angle OPB$$

$$\angle OQD = \angle OBP$$

and $OD = OB$

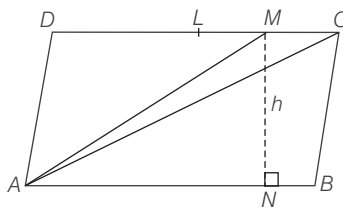
$$\therefore \triangle OQD \cong \triangle OPB \quad [\text{By AAS}]$$

$$\Rightarrow PB = DQ$$



$$\begin{aligned} \text{Now, ar (} \parallel \text{gm } ABCD) &= QR \times AB \\ &= QR \times (AP + PB) = QR \times (AP + DQ) \\ &= 2 \times \text{ar (quadrilateral } APDQ) \end{aligned}$$

42. (a) Draw $MN \perp AB$ and let $MN = h$



$$\begin{aligned} \text{Now, ar (trapezium } ABCM) &= \text{ar } (\triangle AMC) + \text{ar } (\triangle ABC) \\ &\Rightarrow \frac{1}{2}(MC + AB) \times h = 4 + \frac{1}{2} \text{ar} \\ &\quad (\text{parallelogram } ABCD) \\ &\Rightarrow \frac{1}{2} \left(\frac{AB}{4} + AB \right) \times h = 4 + \frac{1}{2} \times AB \times h \\ &\quad \left[\because MC = \frac{AB}{4} \right] \end{aligned}$$

$$\Rightarrow \frac{AB \times h}{8} = 4 \Rightarrow AB \times h = 32$$

$$\Rightarrow \text{ar (} \parallel \text{gm } ABCD) = 32 \text{ cm}^2$$

43. (b) I. All squares are parallelograms.

II. No parallelogram is a trapezium.

III. All squares are rhombuses and also rectangles.

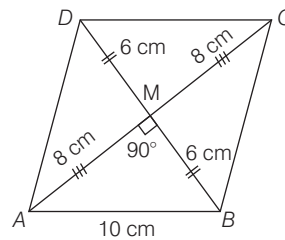
IV. All rhombuses are parallelograms.

So, statements (III) and (IV) are correct.

44. (a) In a rhombus $ABCD$, if AC and BD are two diagonals then

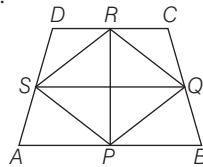
$$\begin{aligned} AB^2 + BC^2 + CD^2 + AD^2 &= AC^2 + BD^2 \end{aligned}$$

$$\begin{aligned} \Rightarrow (10)^2 + (10)^2 + (10)^2 + (10)^2 &= (16)^2 + (12)^2 \\ \Rightarrow 400 &= 400 \end{aligned}$$



Hence, both I and II are true but III is false.

45. (c) $PQRS$ can be shown parallelogram, so the diagonal PR and SQ bisect each other.



II. Area ($\triangle RSQ$)

$$= \frac{1}{2} \text{Area (} \triangle SQCD) \dots(i)$$

$$\text{and area } \triangle (PSQ) = \frac{1}{2} \text{Area (} \triangle PQAS) \dots(ii)$$

On adding Eqs. (i) and (ii), we get

$$\text{Area (} \triangle RSQ) + \text{Area (} \triangle PSQ) = \frac{1}{2}$$

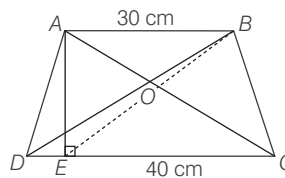
$$[\text{Area (} \triangle SQCD) + \text{Area (} \triangle PQAS)]$$

$$\Rightarrow \text{Area (} \triangle PQRS) = \frac{1}{2} \text{Area (} \triangle ABCD)$$

Hence, both statements are true.

Solutions (Q.Nos. 46-47)

Given, $AO : OC = BO : OD$
and $AB = 30 \text{ cm}$ and $CD = 40 \text{ cm}$



$$\therefore \triangle AOB \sim \triangle COD$$

$$\Rightarrow \frac{OC}{OA} = \frac{AB}{CD} = \frac{30}{40} = \frac{3}{4}$$

$$\therefore \angle OAB = \angle OCD$$

$$\text{and } \angle OBA = \angle ODC$$

It means $DC \parallel AB$.

So, it is a trapezium.

Area of trapezium

$$ABCD = \frac{1}{2} (AB + CD) \times AE$$

$$\Rightarrow 1050 = \frac{1}{2} (30 + 40) \times AE$$

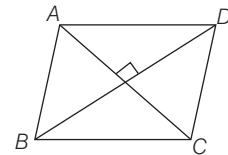
$$\Rightarrow AE = 30 \text{ cm}$$

$$\begin{aligned} 46. (b) \text{ In right } \triangle EAB, EB &= \sqrt{AE^2 + AB^2} \\ &= \sqrt{30^2 + 30^2} = 30\sqrt{2} \text{ cm} \end{aligned}$$

$$\begin{aligned} 47. (c) \text{ Area of } \triangle ADC &= \frac{1}{2} \times CD \times AE \\ &= \frac{1}{2} \times 40 \times 30 = 600 \text{ cm}^2 \end{aligned}$$

48. (c) $ABCD$ is a rhombus.

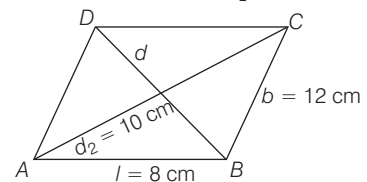
$$\therefore AB = BC = CD = DA$$



and diagonals bisect each other at right angles.

49. (d) In parallelogram,

$$d^2 + d_2^2 = 2(l^2 + b^2)$$



$$\therefore d^2 + (10)^2 = 2(64 + 144)$$

$$\Rightarrow d^2 = 2 \times 208 - 100$$

$$\Rightarrow d^2 = 416 - 100 = 316$$

$$\Rightarrow d = \sqrt{316} \Rightarrow d = 17.76 \text{ cm}$$

$$\therefore d > 12$$

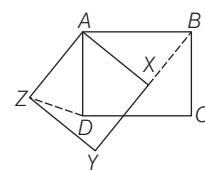
50. (d) In $\triangle ABX$ and $\triangle ADZ$,

$AB = AD$ (side of a square $ABCD$)

and $AX = AZ$ (side of a square $AXYZ$)

Let $\angle BAX = \theta$

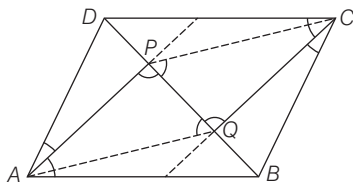
$$\therefore \angle XAD = 90^\circ - \theta$$



Also, $AXYZ$ is a square.

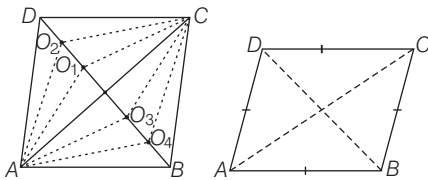
$\therefore \angle ZAX = 90^\circ$
 $\Rightarrow \angle ZAD + \angle XAD = 90^\circ$
 $\Rightarrow \angle ZAD = 90^\circ - (90^\circ - \theta) = \theta$
 i.e. $\angle BAX = \angle ZAD$
 $\therefore \triangle ABX \cong \triangle ADZ$
 $\therefore BX = DZ$ (by CPCT)

- 51. (b)** Since, line segment AP and CQ bisects the $\angle A$ and $\angle C$, respectively.
 Then, $AP \parallel CQ$
 Now, in $\triangle APQ$ and $\triangle CQP$,
 $\therefore AP \parallel CQ$



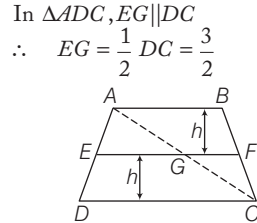
$\therefore \angle APQ = \angle PQC$ (alternate angle)
 $PQ = PQ$ (common)
 Also, $PC \parallel AQ$
 $\therefore \angle CPQ = \angle PQA$ (alternate angle)
 $\therefore \triangle APQ \cong \triangle CQP$ (by ASA)
 Hence, $\triangle APQ$ is similar to $\triangle PCQ$.

- 52. (a)** We know that, in a rhombus $ABCD$ diagonals bisect each other at point O which means that the distance of O from four vertices of rhombus i.e., A, B, C and D are equal. Even, if we take any fixed point on diagonal BD of rhombus $ABCD$ and join with vertices A and C , we get a point which is equidistant from A and C . (by property of congruent triangle). Hence, the locus of a point in rhombus $ABCD$ which is equidistant from A and C is a fixed point on diagonal BD .



- 53. (c)** In parallelogram $PQRS$
 $\angle P + \angle S = 180^\circ$
 $\Rightarrow \frac{1}{2} \angle P + \frac{1}{2} \angle S = 90^\circ$
 $\Rightarrow \angle DPS + \angle DSP = 90^\circ$
 In $\triangle DPS$,
 $\angle SDP = 180^\circ - (\angle DPS + \angle DSP) = 90^\circ$
 $\therefore \angle ADC = \angle SDP = 90^\circ$
 (vertically opposite $\angle S$)
 Similarly,
 $\angle DAB = \angle ABC = \angle BCD = 90^\circ$
 Hence, $ABCD$ is a rectangle.

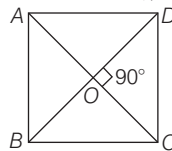
- 54. (c)** Join AC . In $\triangle ACD$, $EG \parallel DC$ and E and G are mid-points of AD and AC , respectively.



Similarly, in $\triangle ABC$
 $GF = \frac{1}{2} AB = 1$
 $EF = EG + GF = 1 + \frac{3}{2} = \frac{5}{2}$

$\therefore$ Area of trapezium = $\frac{1}{2}$
 (sum of parallel sides $\times$ height)
 Now, required ratio = $\frac{\text{Area of } ABFE}{\text{Area of } EFCD}$
 $= \frac{\frac{1}{2} \left(2 + \frac{5}{2} \right) \times h}{\frac{1}{2} \left(3 + \frac{5}{2} \right) \times h} = \frac{9}{11}$ or 9 : 11

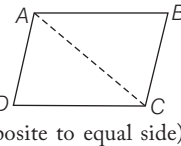
- 55. (b)** Here, $OD = \frac{BD}{2} = \frac{4.8}{2} = 2.4$
 $\Rightarrow OC = \frac{AC}{2} = \frac{1.4}{2} = 0.7$



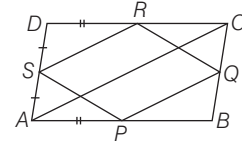
Since, in rhombus diagonal bisect at 90° .
 Then, in $\triangle ODC$, $OD^2 + OC^2 = CD^2$
 $\Rightarrow CD = \sqrt{OD^2 + OC^2}$
 $= \sqrt{(2.4)^2 + (0.7)^2}$
 $CD = \sqrt{6.25} \Rightarrow CD = 2.5 = \frac{5}{2}$

$\therefore$ Perimeter of rhombus
 $= 4a = 4 \times \frac{5}{2} = 10$ cm

- 56. (c)** Join AC .
 Now, in $\triangle ABC$
 $\therefore AB = BC$ (given)
 $\therefore \angle BAC = \angle BCA$
 ... (i)
 (angles opposite to equal side)
 In $\triangle ADC$,
 $\therefore CD > AD$
 $\therefore \angle DAC > \angle DCA$... (ii)
 (since in a triangle, angle opposite to greater side is bigger than the angle opposite to smaller side)
 On adding Eqs. (i) and (ii), we get
 $\angle BAD > \angle BCD$

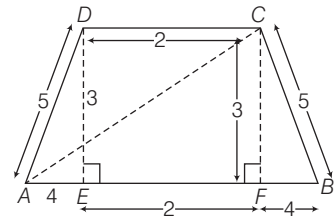


- 57. (b)** In $\triangle APS$ and $\triangle DSR$
 $AP = DR$ [$\because P$ and R are mid points]
 $AS = DS$ [$\because S$ is the mid point]
 and $SP = SR$ [$\because P$ or RS us a]
 $\therefore \triangle APS \cong \triangle DRS$ [By SSS]
 $\Rightarrow$ Area of $\triangle APS$ = Area of $\triangle DSR$



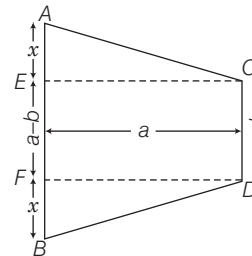
$\therefore AS = SD$ and $AP = DR$
 In $\triangle ABC$ and $\triangle PBQ$
 $\angle ABC = \angle PBQ$ [common]
 $\angle BAC = \angle BPQ$ [$\because PQ \parallel AC$]
 $\therefore \triangle ABC \sim \triangle PBQ$
 $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PBQ)} = \frac{AB^2}{PB^2} \Rightarrow \left[\frac{2PB}{PB} \right]^2$
 $\therefore \text{ar}(\triangle ABC) = 4 \text{ar}(\triangle BPQ)$
 Hence, only statement II is correct.

- 58. (b)**



In $\triangle BCF$,
 By pythagoras theorem,
 $(5)^2 = (3)^2 + (BF)^2 \Rightarrow BF = 4$ cm
 $\therefore AB = 2 + 4 + 4 = 10$ cm
 Now in $\triangle ACF$, $AC^2 = CF^2 + FA^2$
 $\Rightarrow AC^2 = 3^2 + 6^2$, $AC = \sqrt{45}$ cm
 Similarly, $BD = \sqrt{45}$ cm
 $\therefore$ Sum of diagonal = $2 \times \sqrt{45}$
 $= 2 \times 3\sqrt{5} = 6\sqrt{5}$ cm

- 59. (d)** Since, they are symmetrically lying on horizontal plane.



$\therefore AC = BD \therefore AE = BF = x$ (say)
 Now, $AB = (a - b) + 2x$
 i.e. $a + b = a - b + 2x$
 $\Rightarrow 2b = 2x$
 $\therefore x = b$